

# 'Rente mit 63'

## Labour Force Participation among Eligible Older Workers in Germany

*SPS Pareto Seminar - Research Projects in Empirical Economics*

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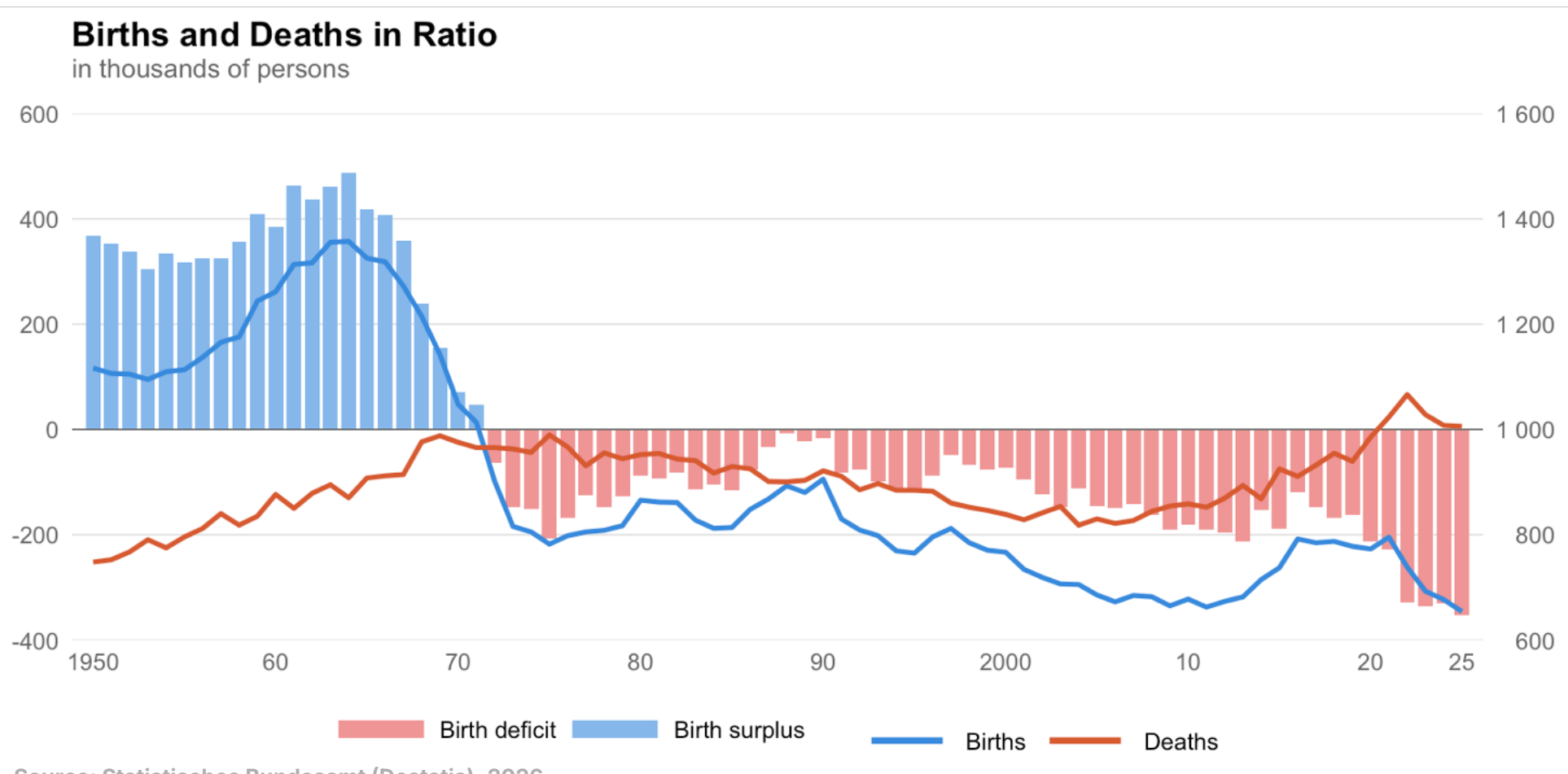
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June 2026

# Germany Is Shrinking

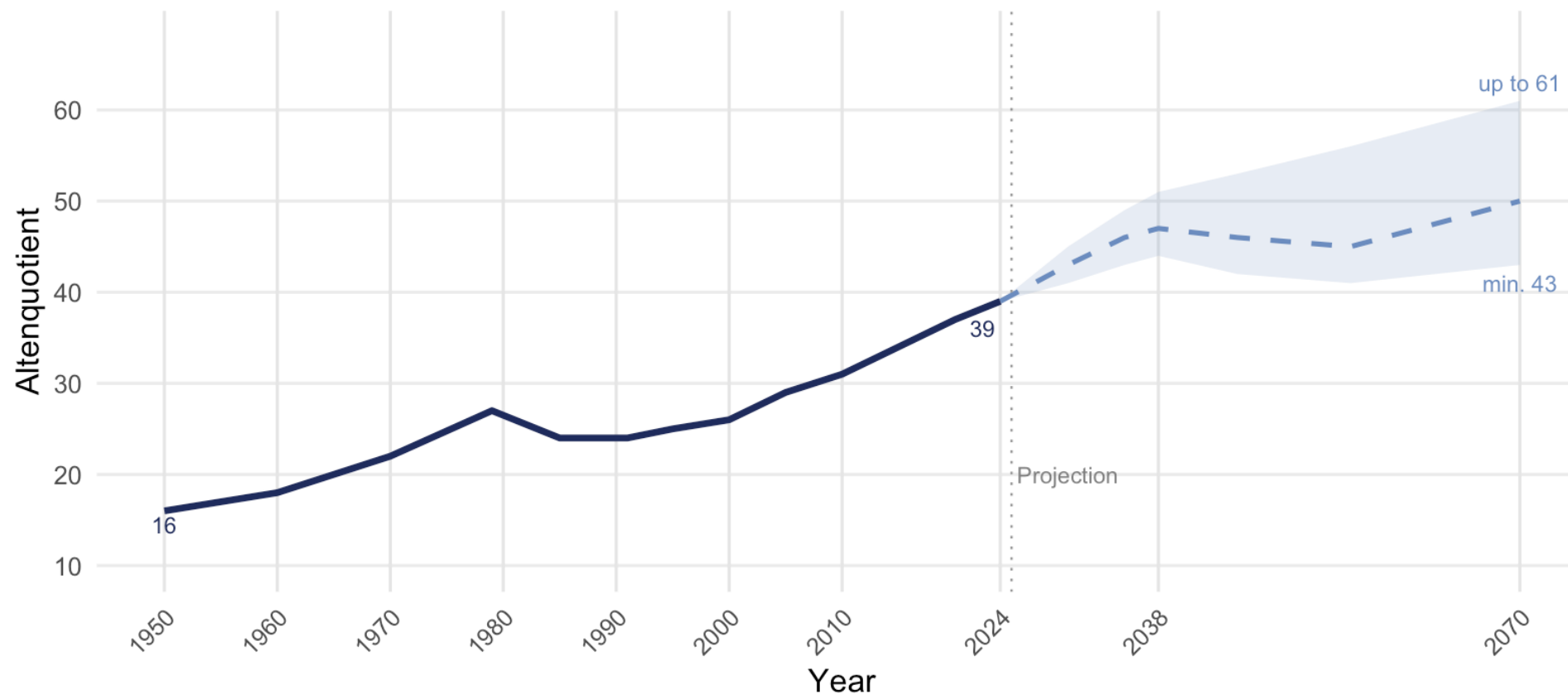


Source: Statistisches Bundesamt (Destatis), 2026

# The Demographic Burden Is Growing Fast

## Old-Age Dependency Ratio in Germany, 1950 to 2070

Persons aged 65+ per 100 persons aged 20-64 | Shaded band: projection range (favourable to unfavourable scenario)



Source: Destatis, 16. koordinierte Bevölkerungsvorausberechnung (2025). Historical values: Destatis (2024).

# How Germany's Pension System Works

- **Pay-as-you-go**
  - *Current workers finance current retirees*
- **Early retirement penalty**
  - *Permanent deduction of 0.3% per month for early retirement*
- **Contribution years as eligibility**
  - *45 qualifying years enable access to early retirement provisions*

# Reform: Rente mit 63

- **Implemented**

- *July 2014*

- **Eligibility**

- *45+ contribution years*

- **Retirement age**

- *reduced to 63, no deductions*

| Birth Year  | Earliest Retirement Age |
|-------------|-------------------------|
| before 1953 | 63 + 0 months           |
| 1953        | 63 + 2 months           |
| 1954        | 63 + 4 months           |
| 1955        | 63 + 6 months           |
| 1956        | 63 + 8 months           |
| ...         | ...                     |
| 1964+       | 67                      |

Source: Deutsche Rentenversicherung Bund

## MAIN RESEARCH QUESTION

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***Did the introduction of Rente mit 63 significantly reduce labour force participation among eligible workers?***

## RQ1 · REGIONAL HETEROGENEITY

### **East vs. West Germany**

*Does the effect vary between East and West Germany?*

## RQ2 · SOCIOECONOMIC HETEROGENEITY

### **RQ2a: Income**

*Does the effect differ across income groups?*

### **RQ2b: Education**

*Does the effect differ across education groups?*

# Difference-in-Differences Framework

## Main Regression

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \epsilon_{it}$$

## RQ1 · Regional Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 (Treat_i \times Post_t \times East_i) + \epsilon_{it}$$



# Difference-in-Differences Framework

## RQ2a · Income Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \sum_{k=2}^5 \gamma_k (Treat_i \times Post_t \times incq_k) + \epsilon_{it}$$

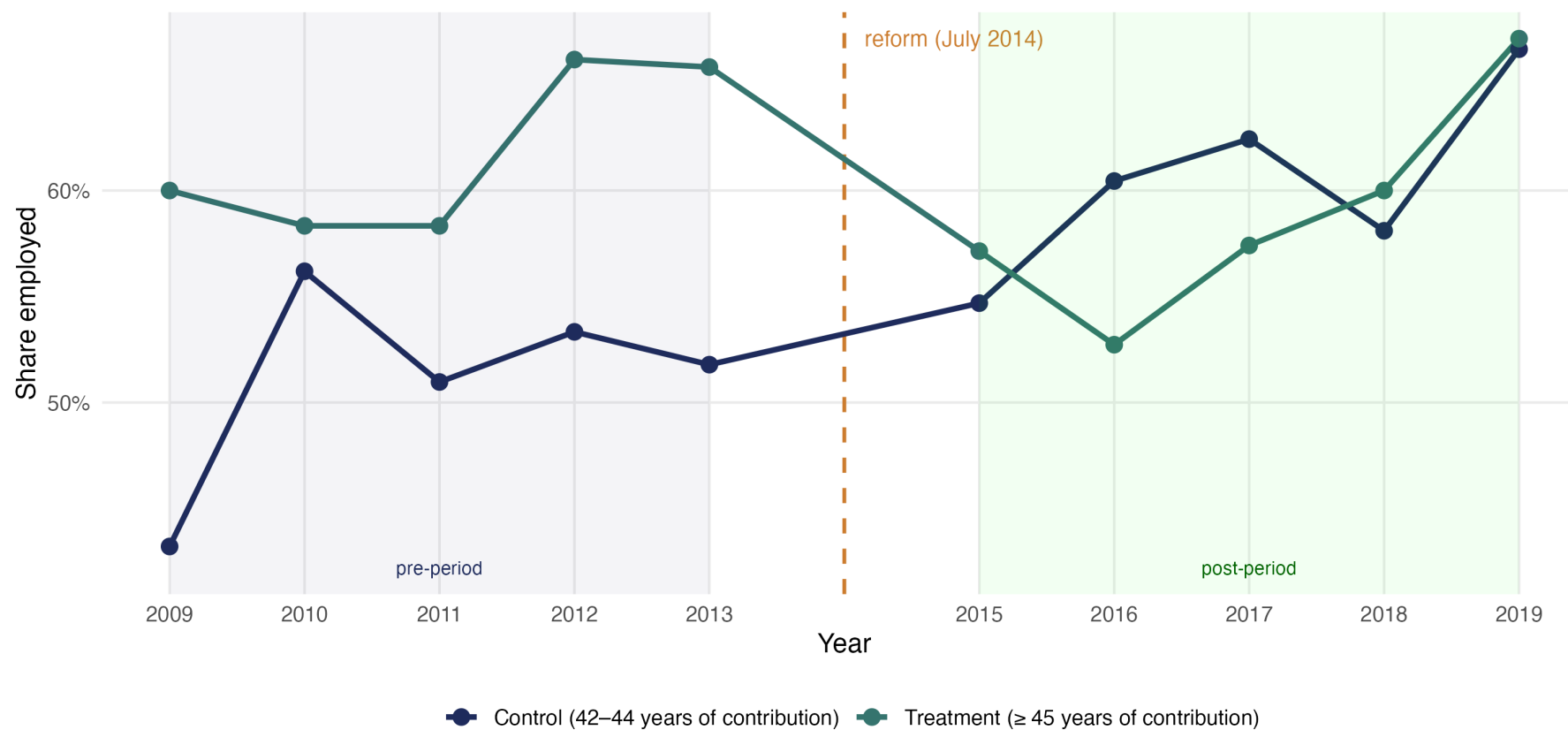
## RQ2b · Educational Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 (Treat_i \times Post_t \times HighEduc_i) + \epsilon_{it}$$

# Control Group Selection

## Employed: treatment- vs. controlgroup (2009–2019)

Control: 42–44 contribution years | 2014 excluded | Parallel Trends:  $p = 0.097$



Source: SOEP-Data

| Main Regression  |                      |
|------------------|----------------------|
| <i>Variables</i> |                      |
| Post             | −0.342***<br>(0.057) |
| Treat × Post     | −0.219***<br>(0.050) |
| Observations     | 1,674                |
| Individual FE    | Yes                  |

\* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001; SOEP-Data

# Insignificant Heterogeneity

|                          | RQ1: East/West       | RQ2b: Education      |
|--------------------------|----------------------|----------------------|
| <i>Variables</i>         |                      |                      |
| Post                     | −0.347***<br>(0.056) | −0.342***<br>(0.057) |
| Treat × Post             | −0.173***<br>(0.052) | −0.212***<br>(0.058) |
| Treat × Post × East      | −0.172<br>(0.111)    |                      |
| Treat × Post × High Educ |                      | −0.016<br>(0.094)    |
| Observations             | 1,674                | 1,674                |
| Individual FE            | Yes                  | Yes                  |

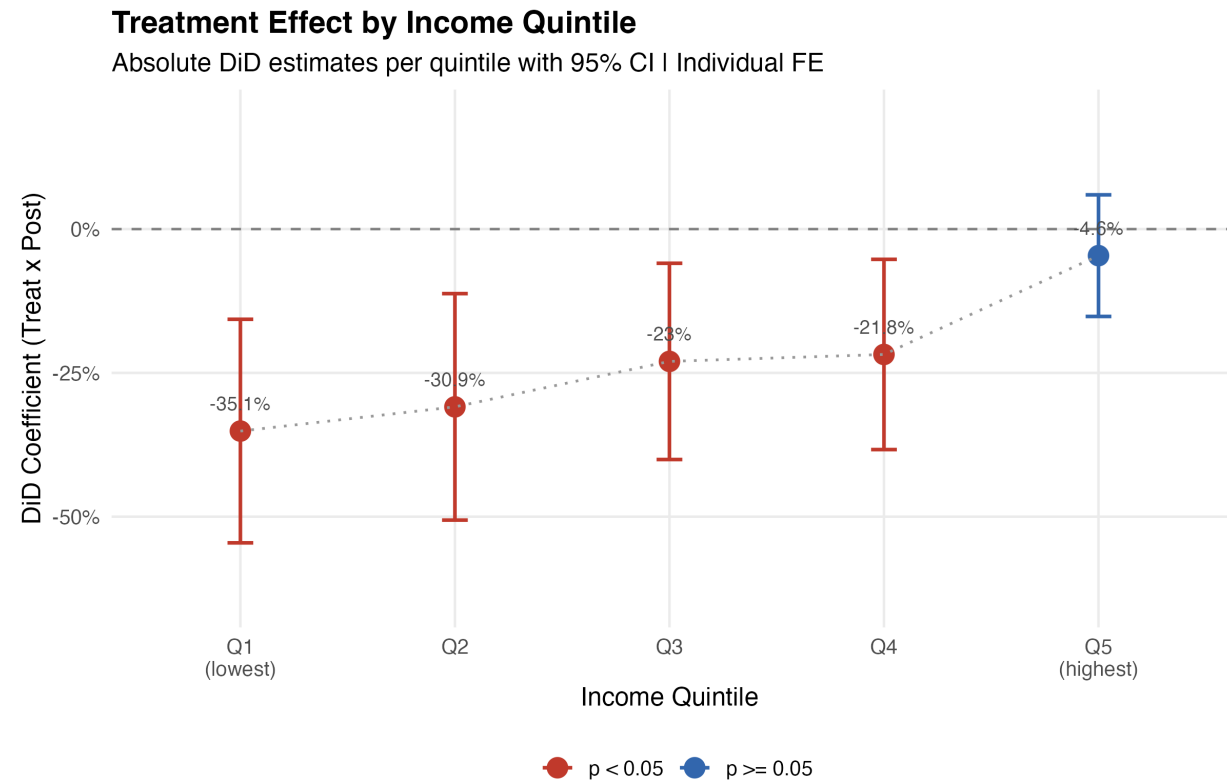
\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ; SOEP-Data

# Significant Heterogeneity

## RQ2a: Income Gradient

| Variables              |                      |
|------------------------|----------------------|
| Post                   | -0.337***<br>(0.057) |
| Treat × Post (Q1 Ref.) | -0.351***<br>(0.099) |
| Treat × Post × Q2      | +0.042<br>(0.114)    |
| Treat × Post × Q3      | +0.121<br>(0.123)    |
| Treat × Post × Q4      | +0.133<br>(0.124)    |
| Treat × Post × Q5      | +0.305**<br>(0.110)  |
| Observations           | 1,674                |
| Individual FE          | Yes                  |

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ; SOEP-Data



Source: SOEP-Data

# Robustness: Alternative Control Groups

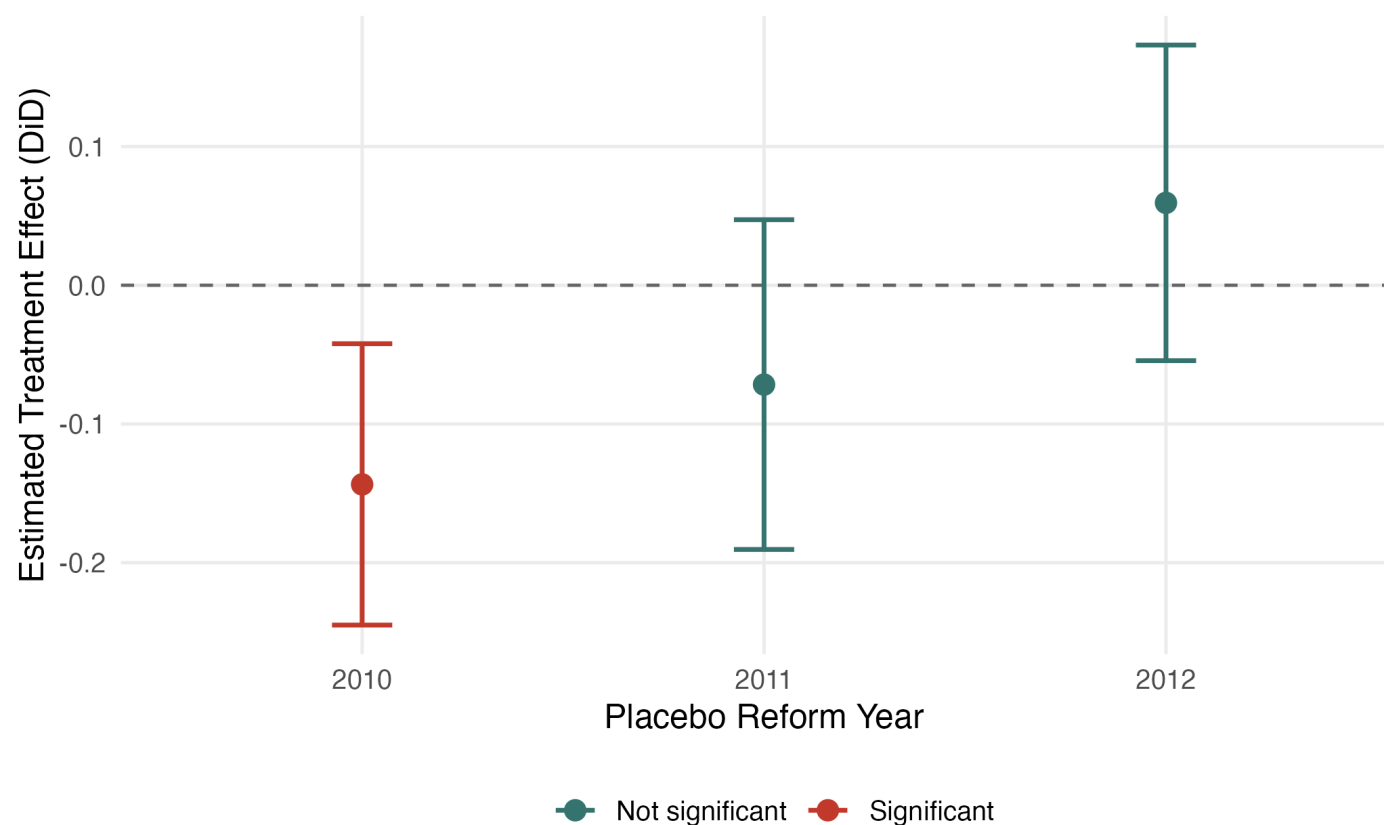
|                  | 35–44                | 40–44               | 41–44               | 42–44*             |
|------------------|----------------------|---------------------|---------------------|--------------------|
| <i>Variables</i> |                      |                     |                     |                    |
| Treat            | 0.239***<br>(0.038)  | 0.170***<br>(0.040) | 0.139***<br>(0.041) | 0.110*<br>(0.043)  |
| Post             | 0.136***<br>(0.018)  | 0.103***<br>(0.025) | 0.108***<br>(0.028) | 0.096**<br>(0.032) |
| Treat × Post     | −0.170***<br>(0.053) | −0.137*<br>(0.055)  | −0.142*<br>(0.056)  | −0.130*<br>(0.058) |
| N (Total)        | 6,553                | 3,438               | 2,731               | 2,068              |
| Ratio (C:T)      | 10.5:1               | 5.0:1               | 3.8:1               | 2.6:1              |

\* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001 ; SOEP-Data

# Robustness: Placebo Tests

## Placebo Test – Estimated DiD Effects

Fake reform years (pre-2014 period only)



Source: SOEP-Data

# Robustness: Donut Design

|                  | Full Sample          | Donut Sample         |
|------------------|----------------------|----------------------|
| <i>Variables</i> |                      |                      |
| Post             | −0.342***<br>(0.057) | −0.339***<br>(0.077) |
| Treat × Post     | −0.219***<br>(0.050) | −0.194**<br>(0.073)  |
| Observations     | 1,674                | 1,232                |
| Individual FE    | Yes                  | Yes                  |

\* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001 ; SOEP-Data



# Robustness: Potential Confounder Flexi-Rente

|                        | Baseline             | Flexi-Dummy          | 2015–16 only         |
|------------------------|----------------------|----------------------|----------------------|
| <i>Variables</i>       |                      |                      |                      |
| Post                   | −0.342***<br>(0.057) | −0.396***<br>(0.057) | −0.410***<br>(0.064) |
| Treat × Post           | −0.219***<br>(0.050) | −0.114*<br>(0.052)   | −0.088<br>(0.079)    |
| Post-2017 (Flexirente) |                      | −0.342***<br>(0.045) |                      |
| Observations           | 1,674                | 1,674                | 1,072                |
| Individual FE          | Yes                  | Yes                  | Yes                  |
| Post-Period            | 2015–19              | 2015–19              | 2015–16              |

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ; SOEP-Data

- **Men only**
  - *Women rarely reach 45 years of contribution*
  
- **Data**
  - *Small treatment group*
  - *Approximated contribution years*
  
- **Identification**
  - *Flexirente effects post-2017*

- Rente mit 63 causally reduces employment
  - *by ~22 percentage points (DiD = -0.219<sup>\*\*\*</sup>)*
- No uniform heterogeneity across subgroups
  - *region and education show no significant differential effects*
- Strong income gradient
  - *low earners (-35 pp) react most, top earners barely affected (-5 pp)*

# Thank You!

for your attention. We look forward to your questions.

SLIDES & PAPER

